# David Xia

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## Research Interests

Reinforcement learning, language model agents, planning and exploration, learning from interaction.

## Education

**University of Illinois Urbana-Champaign** Aug 2023 - May 2027 (expected)
B.S. in Computer Science; B.S. in Mathematics; B.S. in Statistics
Champaign, Illinois · GPA 3.99 / 4.00

## Publications & Papers Under Review

**Designing and benchmarking agentic optimizers**
First author · title withheld during double-blind review
*Under review at ICLR 2027.*

**RSK linear operators and the Vershik-Kerov-Logan-Shepp curve**
Duy Phan, **David Xia**
Authors in alphabetical order, as is standard in mathematics
*Published in Electronic Journal of Combinatorics, 33(3), #P3.17, 2026.* [Journal] [arXiv]

**EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations**
Jung-Woo Chang, Ke Sun, **David Xia**, Xinyu Zhang, Farinaz Koushanfar
*Published at IEEE Symposium on Security and Privacy 2025.* [IEEE] [arXiv]

## Workshop Papers & Manuscripts

**Frozen Policy Iteration for MDPs with Stochastic Transitions under Linear $Q^\pi$ Realizability**
**David Xia**, Ruizhong Qiu, Hanghang Tong
*Manuscript, 2026.* [PDF]

**Predictability in Major League Sports: Betting Odds versus Mathematical Models**
**David Xia**, AJ Hildebrand
*Submitted to Mathematics and Sports (MAS) Journal, 2026.* [PDF]

**Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism**
**David Xia**, Chris Maury
*HEAL Workshop at CHI 2026.* [PDF]

## Research Experience

**Northwestern University** May 2026 - present
Advised by Prof. Zhaoran Wang

- Developed and evaluated language model agents as black-box optimizers, studying how they propose solutions, interpret evaluation results, and revise their search strategies.
- First author of a paper under review at ICLR 2027. Project details withheld during double-blind review.

**iDEA-iSAIL Lab** Feb 2026 - present

Advised by Prof. Hanghang Tong

- Studied sample-efficient exploration and function approximation in reinforcement learning, focusing on Markov decision processes with linearly representable policy values.
- Developed a manuscript extending Frozen Policy Iteration to stochastic transitions under linear $Q^\pi$ realizability, with a sample-complexity guarantee under an additional coverage assumption.
- Currently investigating an algorithm for deterministic MDPs with improved sample-complexity guarantees, while extending the analysis to broader settings.

**Illinois Geometry Lab** Jan 2024 - May 2026

Advised by Prof. AJ Hildebrand

- Led a study of sports predictability, comparing betting odds, Elo ratings, polls, and statistical models. Presented at Rose-Hulman in 2024 and the Joint Mathematics Meetings in 2025.
- Paper submitted to the Mathematics and Sports (MAS) Journal.

**Illinois Combinatorics Lab for Undergraduate Experience (ICLUE)** Aug 2024 - May 2026

Advised by Prof. Alexander Yong

- Proved asymptotic probability bounds for Schensted insertion and RSK linear operators using the Vershik-Kerov-Logan-Shepp limit shape curve. Published in the *Electronic Journal of Combinatorics* in 2026.

**Carnegie Mellon University, HCII** May 2025 - Feb 2026

Advised by Prof. Jeff Bigham

- Built an LLM pipeline for city council reporting, from transcript segmentation to headline generation and topic prioritization.
- Led crowdsourced evaluation of headline quality and topic prioritization. Presented the work at the HEAL workshop at CHI 2026.

**University of California San Diego** Jun 2023 - Dec 2024

Advised by Prof. Xinyu Zhang

- Implemented EveGuard's domain-translation model for estimating eavesdropped audio, enabling end-to-end training of protective audio perturbations with limited data.
- The resulting system achieved over 97% protection against tested audio classifiers while preserving audio quality. Published at IEEE Security and Privacy 2025.

## Teaching

**Course Assistant, CS 374, UIUC** Spring 2026, Fall 2026

## Research Support

**2025** CMU HCII REU participant, supported by NSF award 2349558
**2025** ICLUE RTG participant, supported by NSF award 1937241

## Relevant Coursework

Bold courses are graduate-level.
**In progress (Fall 2026): Statistical Reinforcement Learning, Randomized Algorithms**
**CS:** Machine Learning, Deep Learning, Natural Language Processing, Algorithms, Numerical Analysis, Numerical Methods
**Math:** Honors Real Analysis, **Algebraic Combinatorics**, Abstract Linear Algebra, Combinatorics, Graph Theory, Abstract Algebra, Complex Variables
**Stats:** Bayesian Analysis

## Languages

**English:** Native
**Mandarin Chinese:** Fluent, HSK6